

Iteration and Loop Control Structures in C

> **Definition to Remember**

1. Concept Overview & Fundamentals

Iteration and Loop Control Structures in C forms a core cornerstone of computer systems programming, hardware-level software engineering, and competitive problem solving.

In the C execution model:

2. Technical Architecture & Memory Mechanics

Memory Layout & Storage Organization

```text

### Core Architecture Breakdown

| Component / Concept | Technical Specification | Operational Characteristics | Performance / Complexity |

| :--- | :

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3. Core Syntax & Implementation Patterns

Essential Idioms & Header Declarations

```
```c
// Core standard implementation pattern
int main() {
    // Core logic execution
    printf("Core Logic Executed\n");
    return status;
}
```

4. Real-World Applications & System Engineering

- **Operating System Kernels:** Linux, UNIX, and Windows NT kernels utilize C for drivers, process scheduling, and memory paging.

- **E**

5. Comprehensive Solved Problems & Code Walkthroughs

Problem 1: ArmstrongNumber

Problem Description & Objective:

****C Code Implementation:****

```
#include<stdio.h>
```

```
int num, originalNum, rem, result = 0;
```

```
printf("Enter a number: ");
```

```
originalNum = num;
```

```
while(originalNum != 0) {
```

```
if(result == num)
```

```
return 0;
```

```
---
```

Problem 2: EvenPrint

****Problem Description & Objective:****

****C Code Implementation:****

```
#include<stdio.h>
```

```
int main() {
```

```
int n;
```

```
printf("Enter limit: ");
```

```
scanf("%d", &n);
```

```
printf("Even numbers are:\n");
```

```
for(int i = 1; i <= n; i++) {
```

```
if(i % 2 != 0) {
```

```
continue;
```

```
printf("%d ", i);
```

```
}
```

```
return 0;
```

```
}
```

```
---
```

Problem 3: Factorial

****Problem Description & Objective:****

Write a C program to calculate the factorial of a given number.

****C Code Implementation:****

```
```c
```

```
#include<stdio.h>
```

```
int main()
```

```
printf("Factorial\n\n");
```

```
int num;
```

```
printf("Enter a number: ");
```

```
int fact = 1;
```

```
for(int i = 1; i <= num; i++)
```

```
return 0;
```

```
}
```

```

```

#### Problem 4: Fibonacci

**\*\*Problem Description & Objective:\*\***

Progra

**\*\*C Code Implementation:\*\***

```
```c
```

```
#include<stdio.h>
```

```
int main() {
```

```
/*
```

```
int n, first = 0, second = 1, next;
```

```
printf("Enter limit: ");
```

```
scanf(" %d", &n);
```

```
printf("Fibonacci Series: ");
```

```
while(first <= n) {
```

```
printf("%d ", first);
```

```
next = first + second;
```

```
first = second;
```

```
return 0;
```

```
}
```

```
---
```

Problem 5: GCD

****Problem Description & Objective:****

Create

****C Code Implementation:****

```
```c
```

```
#include <stdio.h>
```

```
int main() {
```

```
/*
```

```
printf("Welcome to GCD Calculator\n");
```

```
printf("Please enter the first number: ");
```

```
if (scan
```

```
printf("Now, enter the second number: ");
```

```
if (scan
```

```
int temp_first = (first < 0) ? -first : first;
```

```
int tem
```

```
while (temp_second != 0) {
```

```
int tem
```

```
int gcd = temp_first;
```

```
printf("The GCD of %d and %d is %d\n", first, second, gcd);
```

```
return 0;
```

```
}
```

```

```

### **Problem 6: InfiniteLoop**

**\*\*Problem Description & Objective:\*\***

3. Prog

**\*\*C Code Implementation:\*\***

```
```c
```

```
#include<stdio.h>
```

```
int main() {
```

```
/*
```

```
int num;
```

```
while(1) {
```

```
printf("Enter a number (-1 to exit): ");
```

```
scanf("%d", &num);
```

```
if(num == -1) {
```

```
break;
```

```
printf("Square = %d\n", num * num);
```

```
}
```

```
printf("Program Ended");
```

```
return 0;
```

```
}
```

```
---
```

Problem 7: LCM

****Problem Description & Objective:****

Create

****C Code Implementation:****

```
```c
```

```
#include <stdio.h>
```

```
int main() {
```

```
/*
```

```
int first, second;
```

```
printf("Welcome to LCM Calculator\n");
```

```
printf("Please enter the first number: ");
```

```
scanf("%d", &first);
```

```
printf("Now, enter the second number: ");
```

```
scanf("%d", &second);
```

```
int min = (first > second) ? first : second;
```

```
int max = first * second;
```

```
for (int i = min; i <= max; i++) {
```

```
if (i % first == 0 && i % second == 0) {
```

```
printf("The LCM of %d and %d is %d\n", first, second, i);
```

```
break;
```

```
}
```

```
return 0;
```

```
}
```

```

```

### Problem 8: Multiplication

**\*\*Problem Description & Objective:\*\***

Develop a C program to print the multiplication table of a given number.

**\*\*C Code Implementation:\*\***

```
```c
```

```
#include <stdio.h>
```

```
int main()
```

```
printf("Multiplication Table\n\n");
```

```
int num;
```

```
printf("Enter number: ");
```

```
scanf("%d", &num);
```

```
for (int i = 1; i <= 12; i++){
```

```
printf("%d x %d = %d\n", num, i, (num * i));
```

```
}
```

Problem 9: OddSum

****Problem Description & Objective:****

****C Code Implementation:****

```
#include <stdio.h>
```

```
int main() {
```

```
int sum = 0;
```

```
printf("Enter value of N: ");
```

```
for (int i = 1; i <= num; i++) {
```

```
return 0;
```

Problem 10: PalindromeNumber

****Problem Description & Objective:****

****C Code Implementation:****

```
#include<stdio.h>
```

```
int num, originalNum, reverse = 0, rem;
```

```
printf("Enter a number: ");
```

```
scanf("%d", &num);
```

```
originalNum = num;
```

```
while(num != 0) {
```

```
    rem = num % 10;
```

```
    if(originalNum == reverse)
```

```
        printf("Palindrome\n");
```

```
    return 0;
```

```
}
```

```
---
```

Problem 11: PatternPrint

****Problem Description & Objective:****

Implement the following pattern:

****C Code Implementation:****

```
```c
```

```
int main() {
```

```
 /*
```

```
 int rows, i, j;
```

```
 printf("Enter number of rows: ");
```

```
 scanf("%d", &rows);
```

```
 // Right Half Pyramid
```

```
printf("\n\n");

for(i = 1; i <= rows; i++) {

 // Reverse Right Half Pyramid

 for(i = rows; i >= 1; i--) {

 // Left Half Pyramid

 for(i = 1; i <= rows; i++) {

 for(j = 1; j <= i; j++) {

 printf("%d\t", i-j+1);

 if(j % 10 == 0) printf("\n");

 }

 printf("\n");

 }

 }

}

return 0;

```

### Problem 12: PrimeNumber

**\*\*Problem Description & Objective:\*\***

Progra

**\*\*C Code Implementation:\*\***

```c

```
#include<stdio.h>
```

```
int main() {
```

```
/*
```

```
int num, i = 2, isPrime = 1;
```

```
printf("Enter a number: ");
```

```
if(num <= 1) {
```

```
while(i < num) {
```

```
if(isPrime)
```

```
return 0;
```

```
---
```

Problem 13: ReverseNumber

****Problem Description & Objective:****

****C Code Implementation:****

```
#include<stdio.h>
```

```
int main() {
```

```
int num, reverse = 0, rem;
```

```
printf("Enter a number: ");
```

```
while(num != 0) {
```

```
rem =  
  
printf("Reversed Number = %d", reverse);  
  
return 0;  
}  
  
---
```

Problem 14: SumOfDigit

****Problem Description & Objective:****

Create

****C Code Implementation:****

```c

```
#include <stdio.h>
```

```
int main() {
```

/\*

```
printf("Sum of Digits!!!\n\n");
```

```
int num, digit, sum = 0;
```

```
printf("Enter number: ");
```

scanf(

```
while (num != 0) {
```

```
digit = num % 10;
```

```
sum += digit;
```

```
num = num / 10;
```

}

```
printf("Sum of digits = %d\n", sum);
```

```
return 0;
```

}

```

```

### Problem 15: SumPositive

**\*\*Problem Description & Objective:\*\***

1. Prog

**\*\*C Code Implementation:\*\***

```c

```
#include<stdio.h>
```

```
int main() {
```

/*

```
int num, sum = 0;
```

```
printf("Enter 5 numbers:\n");
```

```
for(int i = 1; i <= 5; i++) {
```

```
scanf("%d", &num);
```

```
if(num < 0) {
```

contin

```
sum = sum + num;
```

}

```
printf("Sum of positive numbers = %d", sum);
```

```
return 0;
```

}

```
---
```

Problem 16: even numbers continue

****Problem Description & Objective:****

Implem

****C Code Implementation:****

```c

```
#include <stdio.h>
```

```
int main() {
```

/\*

```
int i;

printf("Even numbers from 1 to 20:\n");

for(i = 1; i <= 20; i++) {

 // Skip odd numbers
```

if(i % 2

```
printf("%d ", i);
```

}

```
return 0;
```

}

```

```

### Problem 17: multiplication table

**\*\*Problem Description & Objective:\*\***

Implemen

**\*\*C Code Implementation:\*\***

```c

```
#include <stdio.h>
```

```
int main() {
```

/*

```
int num, i;
```

```
printf("Enter a number: ");
```

scanf(

```
printf("\nMultiplication Table of %d:\n", num);
```

```
for(i = 1; i <= 10; i++) {
```

```
printf("
```

```
return 0;
```

```
}
```

```
---
```

Problem 18: positive number

****Problem Description & Objective:****

Implemen

****C Code Implementation:****

```
```c
```

```
#include <stdio.h>
```

```
int main() {
```

```
/*
```

```
int num;
```

```
do {
```

```
printf("
```

```
if(num <= 0) {
```

```
printf("
```

```
} while(num <= 0);
```

```
printf("You entered a valid positive number: %d\n", num);
```

```
return 0;
```

```
}
```

```

```

### Problem 19: prime number

**\*\*Problem Description & Objective:\*\***

Implemen

**\*\*C Code Implementation:\*\***

```c

```
#include <stdio.h>
```

```
int main() {
```

```
/*
```

```
printf("Enter a number: ");
```

```
if (scan
```

```
if(num <= 1) {
```

```
isPrim
```

```
for(i = 2; i * i <= num; i++) {
```

```
if(num
```

```
if(isPrime == 1) {
```

```
printf("
```

```
return 0;
```

```
}
```

```
---
```

Problem 20: square break loop****Problem Description & Objective:********C Code Implementation:****

Use an infinite loop and a break statement to exit

```
#include <stdio.h>
```

```
int main() {
```

```
int num;
```

```
while(1) {
```

```
printf("Enter a number (-1 to exit): ");
```

```
// Exit condition
```

```
printf("Square = %d\n", num * num);
```

```
printf("Program Ended.\n");
```

```
return 0;
```

```
---
```

Problem 21: sum positive continue****Problem Description & Objective:********C Code Implementation:****

```
#include <stdio.h>
```

```
int main() {
```

```
/*
```

```
int num, sum = 0;
```

```
printf("Enter numbers (0 to stop):\n");
```

```
while(1) {
```

```
scanf("%d", &num);
```

```
// Stop the loop
```

```
if(num == 0) break;
```

```
// Skip negative numbers
```

```
if(num < 0) continue;
```

```
sum = sum + num;
```

```
}
```

```
printf("Sum of positive numbers = %d\n", sum);
```

```
return 0;
```

```
}
```

```
---
```

Problem 22: sum until zero

****Problem Description & Objective:****

Implement the following program in C.

****C Code Implementation:****

```
```c
```

```
#include <stdio.h>
```

```
int main() {
```

```
/*
```

```
int num, sum = 0;
```

```
do {
```

```
printf("
```

```
sum = sum + num;
```

```
} while(num != 0);
```

```
printf("Total Sum = %d\n", sum);
```

```
return 0;
```

```
}
```

```

```

## 6. Exam Scoring Points & Memory Revision

> **Must-Write Points**

> 1. fo

> **Quick Recall**

> `Init